MODULE 3: AIRCRAFT STABILITY AND CONTROL

ASYS 245: Aircraft Systems I | Cornell Notes – Section 1 of 2

CUES / QUESTIONS

1. What are the 3 axes of flight and what movements do they control?
2. Which primary flight control controls which movement?
3. What is stability and how is it related to each axis?
4. What is a dihedral wing and how does it help with lateral stability?
5. What is the centre of gravity and what happens if it is too far forward or aft?
6. What is the difference between static and dynamic stability?

NOTES

1. THE 3 AXES OF FLIGHT

- Roll – Longitudinal axis (wing tip up/down)
- Pitch – Lateral axis (nose up/down)
- Yaw – Vertical axis (left/right)



2. PRIMARY FLIGHT CONTROLS



Aileron – roll



↑ down
↓ up

Elevator – pitch



↑ up
↓ down

Rudder – yaw



← left
→ right

3. STABILITY

- Lateral stability – stable in roll (longitudinal axis)
- Longitudinal stability – stable in pitch (lateral axis)
- Directional stability – stable in yaw (vertical axis)

STABILITY BY AXIS

Axis	Stability Type
Longitudinal (roll)	Lateral
Lateral (pitch)	Longitudinal
Vertical (yaw)	Directional

4. DIHEDRAL WING & LATERAL STABILITY



Dihedral angle
produce more lift in a sideslip, causing the aircraft to right itself.

STABILITY BY AXIS

Axis	Stability Type
Longitudinal (roll)	Lateral
Lateral (pitch)	Longitudinal
Vertical (yaw)	Directional

5. CENTRE OF GRAVITY (CG)



Too far forward
(Nose heavy)
- Higher stall speed

Within limits
(OK)

Too far aft
(Tail heavy)
- Unstable in pitch

- Helps the lower wing produce more lift during sideslips.

6. STATIC vs DYNAMIC STABILITY



Type	Meaning
Positive	Returns to equilibrium over time.
Neutral	Stays displaced or oscillates.
Negative	Moves further away or oscillates.

SUMMARY – SECTION 1

- The 3 axes (roll, pitch, yaw) control aircraft movement and are linked to the primary flight controls.
- Stability is about the aircraft's tendency to return to or resist a change, not the axis itself.
- Dihedral, proper CG and correct loading all contribute to stability.
- Static stability is the initial response; dynamic stability is the behaviour over time.